721. CLINICAL ALLOGENEIC TRANSPLANTATION: CONDITIONING REGIMENS, ENGRAFTMENT, AND ACUTE TRANSPLANT TOXICITIES: POSTER I | DECEMBER 02, 2016

Oral Vancomycin Is Highly Effective in Preventing *Clostridium Difficile* infection in Allogeneic Hematopoietic Stem Cell Transplant Recipients

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Abstract

Introduction: Infection remains a leading cause of morbidity and mortality following allogeneic hematopoietic stem cell transplantation (alloHCT). Among the most common infectious complications in alloHCT recipients is *Clostridium difficile* infection (CDI), a healthcare-associated, toxin-mediated diarrheal disease occurring in up to 30% of alloHCT recipients. We sought to evaluate whether prophylactic oral vancomycin reduces the incidence of CDI in alloHCT recipients.

Methods: We conducted a retrospective cohort study to examine the effectiveness of CDI prophylaxis with oral vancomycin compared to no prophylaxis in 105 consecutive adults undergoing alloHCT at the University of Pennsylvania between April 2015 and July 2016. We initiated a pilot program of CDI prophylaxis with oral vancomycin for all alloHCT recipients starting in December 2015 (N=50). Patients received oral vancomycin 125 mg twice daily starting on the day of inpatient admission for alloHCT and continued until day of discharge. Prior to the initiation of this pilot, pharmacologic prophylaxis for CDI was

not administered (control arm; N=55). Testing for *C. difficile* in patients with diarrhea was performed using an immunoassay for *glutamate dehydrogenase and toxins A and B*. Indeterminate results were confirmed by a molecular assay. Control and intervention patients were characterized by potential risk factors for CDI, including demographics, prior CDI, and recent broad-spectrum antibiotic use. Continuous variables were compared using the Wilcoxon rank-sum test, and categorical variables were compared using the χ^2 or Fisher exact test. The primary outcome of interest was the incidence of CDI from inpatient admission for alloHCT to discharge.

Results: The rate of recent broad-spectrum antibiotic use was significantly greater in the patients that received oral vancomycin prophylaxis compared to control patients (84.0% vs. 67.2; *P*=0.047). In addition, more patients in the vancomycin prophylaxis group received myeloablative conditioning (62.0% vs. 45.4%; *P*=0.09). There were no other significant differences in patient and transplant characteristics between the intervention and control groups, including mean age (54.3 vs. 56.4 years; *P*=0.6), myeloid malignancy (80.0% vs. 81.8%; *P*=0.8) and history of CDI in the prior year (22.0% vs. 9.0%; *P*=0.1). There were no cases of CDI in patients that received oral vancomycin prophylaxis (0/50; 0%) whereas 11/55 (20%) patients who did not receive vancomycin prophylaxis developed CDI during alloHCT (*P*=<0.001). The median (interquartile range) time to CDI diagnosis was 8 days (6 - 12). The median length of stay was non-significantly longer in the intervention patients (33.5 vs. 28.0 days; *P*=0.06), possibly due to the higher number of myeloablative conditioning transplants in the prophylactic vancomycin group. There were no cases of vancomycin-resistant enterococcus bloodstream infection in patients who received vancomycin prophylaxis.

Conclusion: Prophylactic vancomycin is highly effective in preventing CDI in alloHCT recipients. Since CDI and/or its treatment may alter the gut microbiome, longer follow up will determine if there is any impact on other outcomes. Six month follow-up data for graft-versus-host disease and overall survival will be presented.

Disclosures

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Author notes

*Asterisk with author names denotes non-ASH members.